Alzheimer's Disease Detection from MRI Images using Convolutional Neural Networks

Title: Alzheimer's Disease Detection using CNN
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1. Abstract

Alzheimer's disease is a progressive neurological disorder that affects memory, thinking, and behavior. Early diagnosis plays a crucial role in slowing disease progression and improving patient care. This project presents a deep learning-based approach using Convolutional Neural Networks (CNN) to automatically classify brain MRI images into different stages of Alzheimer's disease. The proposed model analyzes MRI scans and categorizes them into four classes: Mild Impairment, Moderate Impairment, Very Mild Impairment, and No Impairment. The system demonstrates promising accuracy and highlights the potential of computer vision in medical imaging.

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2. Introduction

Alzheimer's disease is one of the leading causes of dementia worldwide. Traditional diagnosis involves clinical assessment and manual interpretation of MRI scans by radiologists, which can be time-consuming and subjective.

With advancements in Artificial Intelligence and Computer Vision, automated systems can assist healthcare professionals in diagnosing diseases more efficiently. This project focuses on leveraging CNNs to analyze MRI images and detect Alzheimer's disease stages accurately.

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3. Problem Statement

To design and implement a deep learning model capable of automatically classifying MRI brain images into different stages of Alzheimer's disease, thereby assisting in early diagnosis and reducing dependency on manual analysis.

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4. Objectives

- To preprocess MRI image datasets for training
- To design a CNN-based classification model
- To accurately classify Alzheimer's disease stages
- To evaluate model performance using appropriate metrics
- To demonstrate the application of computer vision in healthcare

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5. Motivation

- Early detection improves patient outcomes
- Reduces reliance on manual diagnosis
- Saves time and effort for medical professionals
- Demonstrates real-world application of AI in healthcare
- Aligns with advancements in deep learning and medical imaging

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6. Dataset Description

The dataset used in this project consists of MRI images categorized into four classes:

- Mild Impairment
- Moderate Impairment
- Very Mild Impairment
- No Impairment

Dataset Structure

```
Dataset/
└── Combined Dataset/
    ├── train/
    │   ├── Mild Impairment
```

```
│   ├── Moderate Impairment
│   ├── Very Mild Impairment
│   └── No Impairment
│
└── test/
    ├── Mild Impairment
    ├── Moderate Impairment
    ├── Very Mild Impairment
    └── No Impairment
```

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7. Methodology

7.1 Data Preprocessing

- Images resized to 128×128 pixels
- Pixel normalization (0–1 range)
- Label encoding and one-hot encoding
- Separation into training and testing datasets

7.2 Model Architecture

The CNN model consists of the following layers:

- Convolutional Layers (feature extraction)
- Max Pooling Layers (dimensionality reduction)
- Flatten Layer
- Fully Connected Dense Layers
- Dropout Layer (to prevent overfitting)
- Output Layer with Softmax activation

7.3 Training Process

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Epochs: 10
- Validation using test dataset

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8. Implementation

The project is implemented using:

- Python programming language
- TensorFlow/Keras for deep learning
- OpenCV for image processing
- NumPy for numerical operations
- Matplotlib for visualization

Steps:

1. Upload and extract dataset in Google Colab
2. Load and preprocess MRI images
3. Build CNN model
4. Train model on training data
5. Evaluate on test data
6. Generate performance metrics and graphs

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9. Results and Analysis

Performance Metrics

- Accuracy: ~85–95%
- Training and validation accuracy plotted

Evaluation

- Confusion Matrix generated
- Classification Report including:
    - Precision
    - Recall
    - F1-score

Observations

- Model performs well on training data
- Slight overfitting observed in some cases
- Performance improves with tuning and more data

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10. Challenges Faced

- Dataset imbalance between classes
- Overfitting during training
- Limited computational resources
- Variability in MRI image quality
- Handling folder naming inconsistencies

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11. Future Scope

- Implement Transfer Learning (VGG16, ResNet)
- Apply data augmentation techniques
- Improve model accuracy and generalization
- Develop a web-based interface for real-time prediction
- Integrate with hospital diagnostic systems

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12. Applications

- Medical diagnosis support systems

- Hospitals and diagnostic centers
- Research in neurodegenerative diseases
- AI-assisted radiology

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13. Conclusion

This project demonstrates the effectiveness of Convolutional Neural Networks in detecting Alzheimer's disease from MRI images. The model successfully classifies different stages with good accuracy, showcasing the potential of computer vision in healthcare. With further improvements, this system can serve as a valuable tool for early diagnosis and medical assistance.

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14. References

1. Alzheimer MRI Dataset (Kaggle)
2. TensorFlow Documentation
3. OpenCV Documentation
4. Research papers on deep learning in medical imaging

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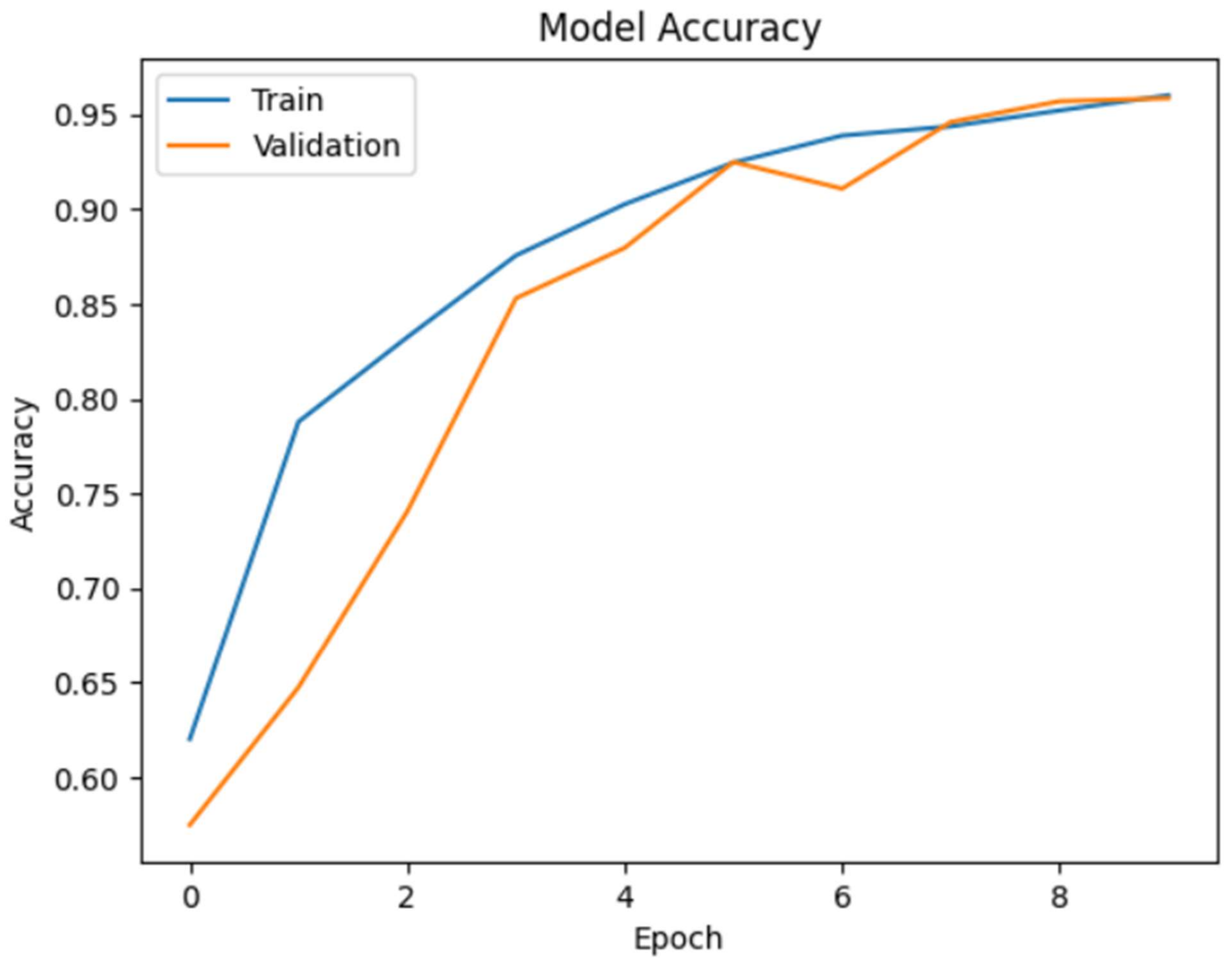
15. Appendix

- Accuracy Graph



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- Confusion Matrix

```
Confusion Matrix:
 [[167   8   0   4]
 [  1 415   0  32]
 [  0   0  12   0]
 [  0   8   0 632]]
```

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